HOUSE BOOK • SHARE EDITION • 2026-10-01

17 Maple Court

Anytown, USA 00000

Two-story colonial · 4 bed / 2.5 bath · built 1994

Systems · panels · quirks · providers · warranties · invoices · paperwork

June 2018 — October 2026



Compiled from email, messages, documents, photos & field notes.

ILLUSTRATION · FICTIONAL HOUSE

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1 / At a Glance

INVESTED SINCE 2018
$82,050
12 projects · best estimate

PAPERWORK ON FILE
8/12
3 partial · 0 missing

WARRANTIES
5 active
2 expiring <1 yr · 4 expired · 1 unknown

OPEN ISSUES
9
2 safety

SERVICE PROVIDERS
12
with contacts

SERVICE LOG
29
visits & repairs, 2018–26

This book is the house's documentation, generated from data: every system, circuit, repair, warranty and invoice, each with a pointer to where the proof lives. It exists so that the next furnace call, insurance claim, contractor bid or sale starts from facts instead of memory. **17 Maple Court is fictional**: it's the example that ships with the engine.

Address	17 Maple Court, Anytown · cul-de-sac lot, house faces south · attached 2-car garage on the east
Built	1994 (tract builder, 3rd phase of the subdivision) · 2×6 exterior walls, vinyl siding over OSB · full unfinished basement, poured walls
Size	~2,450 sq ft above grade (1st 1,250 + 2nd 1,200) · basement ~1,150 sq ft unfinished · lot ~0.31 acre
Bought	2018-06-15 from the original owners (24 years in the house); seller's info sheet + 2018 inspection are the baseline for everything here
HVAC	Two independent zones, both gas furnace + split A/C: zone 1 = basement furnace serving 1st floor + basement; zone 2 = attic furnace serving the 2nd floor (replaced 2020)
Electric	200 A overhead service · Square D Homeline 40-space main panel in the basement · 22 kW standby generator → 100 A automatic transfer switch → 12-circuit generator subpanel
Water	Municipal water + sewer · 1 in copper service, shutoff on the basement north wall · 40-gal gas power-vent water heater (2025) · sump pit with primary + battery-backup pumps
Roof	Architectural asphalt shingles, full tear-off 2019 · ridge vent + soffit intake · 6 in aluminum gutters

In an emergency

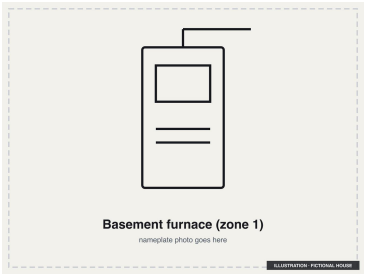
Water leak	Main shutoff: basement north wall, behind the water heater, red-handled ball valve (quarter turn). Then Dry Basin Plumbing 555-0142.
Basement water	Check the sump pit (basement SE corner). Primary pump on generator circuit G-5; the battery backup alarms on its own. If both are running and losing, call Dry Basin and start a wet-vac.
Gas smell	Leave the house, then call the gas utility and 911 from outside. Meter + shutoff on the west wall by the AC pad.
No heat / no cool	Thermostat batteries → furnace switch (light switch on the furnace) → filter → breaker (furnaces are on the generator subpanel: G-1 zone 1, G-3 zone 2). Then Acme Heating & Cooling 555-0117 with zone number + model from the Systems chapter.
Power out	Generator starts in ~10 s and carries only the generator subpanel (see the panel directory). EV charger, both A/C condensers and the kitchen countertops are NOT backed.
Fire / CO	911. Interconnected smoke/CO on every floor (2-yr-old units, replaced 2024). The power-vent water heater and both furnaces share the basement mechanical area.

2 / Systems

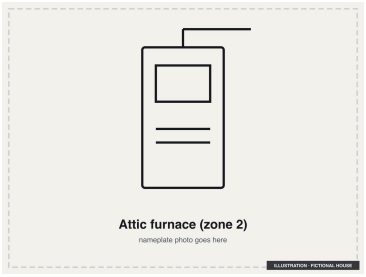
2.1. Heating & cooling

Two independent zones, each a gas furnace with a split A/C condenser on the east side yard. Zone 2 was replaced in 2020; zone 1 is the prior owner's 2012 equipment and the next big-ticket item.

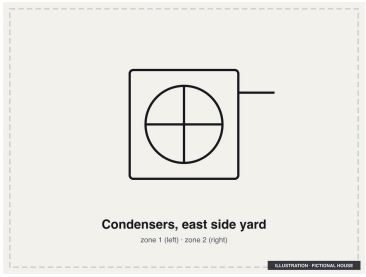
Zone 1 (1st floor + basement)	Carrier 59TP5A080 80k BTU/h 96% furnace in the basement + 24ACC642 condenser, 3.5 t (nameplate) · installed 2012 by the prior owner · the seller's info sheet says "4-ton A/C, 100k furnace": both wrong, the nameplates win · furnace on G-1 (generator), condenser on M-5/7
Zone 2 (2nd floor)	Carrier 59SC5A060 60k BTU/h furnace in the attic + 24ACC624 condenser, 2 t · installed 2020-04-22 by Acme Heating & Cooling · secondary drain pan with float cutoff · furnace on G-3 (generator), condenser on M-9/11
Refrigerant	Both R-410A. Zone 1 has been topped off twice (2024-07: 3 lb + UV dye; 2026-06: 2 lb). The 30-day dye recheck after the first top-off never happened. Treat zone 1 as a known leaker until the coil is inspected.
Filters	Zone 1: 16×25×4 media cabinet (6-month) · zone 2: 20×20×1 at the attic return grille (90-day, it's reachable from the bedroom 4 closet hatch)
Thermostats	Two programmable Wi-Fi thermostats, hall (zone 1) and upstairs landing (zone 2); C-wire present at both
Service	Acme Heating & Cooling: fall tune-up of both zones each October since 2018 (see the invoice ledger)



Zone 1 furnace, basement



Zone 2 furnace, attic (2020)

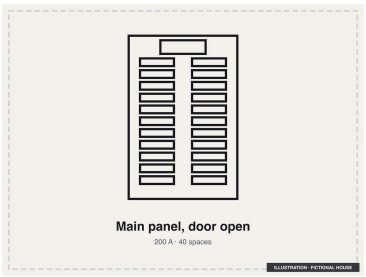


Condensers, east side yard

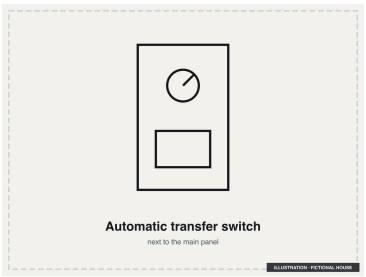
2.2. Electrical & standby power

One 200 A service with a standby generator behind an automatic transfer switch. The full circuit directory, with per-leg load tallies and a demand estimate, is in the Electrical Panels chapter; printable door labels are in the appendix.

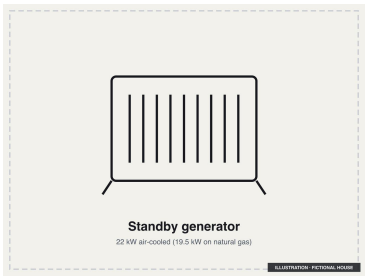
Service	200 A, overhead drop on the east gable · meter on the garage wall · Square D Homeline 40-space main breaker panel (copper bus), basement east wall
Generator	Generac Guardian 22 kW air-cooled (19.5 kW on natural gas), north side yard · 100 A automatic transfer switch next to the main panel · weekly 5-minute exercise, Tuesdays 1 p.m. · annual service by GenReady Power Systems each March
On generator	Both furnaces, sump pump, refrigerator, water heater power vent, laundry, garage door, hall + kitchen lights, primary bedroom, radon fan, basement dehumidifier
NOT on generator	Both A/C condensers, EV charger, kitchen countertops, dishwasher, microwave, all other bedrooms
EV charging	50 A 2-pole circuit (M-2/4) to a hardwired 40 A wall unit in the garage, 2023



Main panel, door open



Transfer switch



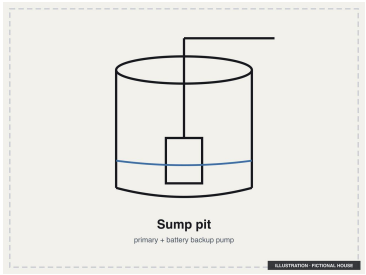
Generator, north side

2.3. Plumbing, water & air

Water heater	Bradford White 40-gal natural-gas POWER-VENT (2025-02, Dry Basin Plumbing) · the vent blower needs 120 V: it's on generator circuit G-9, so hot water survives an outage
Sump	Zoeller M53 primary (2021-05) + Basement Watchdog battery backup with its own alarm · primary on G-5 · discharge to the back yard, 10 ft past the patio · backup battery replaced 2025-10 (5-yr cadence)
Water service	1 in copper from the street, meter + shutoff on the basement north wall · outdoor hose bibs are frost-free (2019)
Radon	Sub-slab depressurization, Riverbend Radon Services 2018-08 · fan in the garage attic on G-10 · manometer on the stack in the basement; post-mitigation test 0.9 pCi/L
Dehumidifier	Basement, 70-pint, drains to the sump pit · on G-12



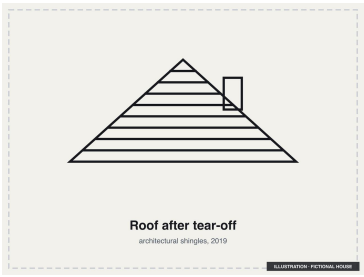
Water heater (2025)



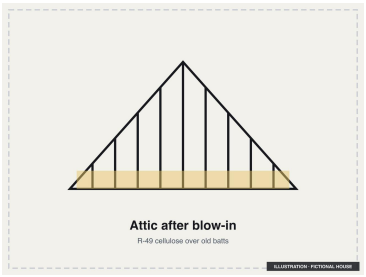
Sump pit

2.4. Envelope: roof, windows, attic, outside

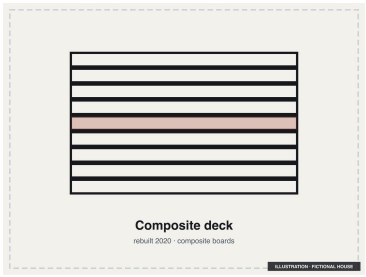
Roof	Full tear-off 2019-05, Summit Roofing Co.: architectural shingles, synthetic underlayment, ice & water shield at eaves + valleys, new step flashing at the garage wall · manufacturer system warranty requires contractor registration (not confirmed, see Paperwork)
Attic	Air-sealed top plates + blown cellulose to R-49 (2019-10, Northwind Insulation) over the original R-19 batts · baffles at every soffit bay
Windows	Original 1994 double-hung vinyl, insulated glass · dining room sashes 2 + 3 re-glazed 2023-06 after seal failure (fogging) · 6 other units show early fogging
Deck	Composite boards on pressure-treated framing, rebuilt 2020 (Deckworks Outdoor); original footings retained and inspected
Patio	380 sq ft concrete pavers, 2024-04 (Greenway Hardscapes) · 3-yr installer warranty covers settling; a low spot by the NE downspout appeared 2026-05



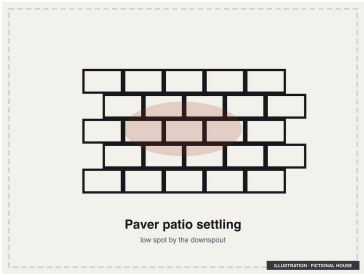
Roof after the 2019 tear-off



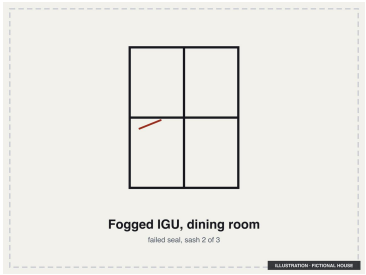
Attic after the 2019 blow-in



Composite deck (2020)



Patio settling, 2026-05



Fogged insulated glass, dining room (2023)

3 / Electrical Panels

One 200 A Square D panel (**M**) and a 12-space generator subpanel (**G**) behind a 100 A automatic transfer switch, fed from M's 60 A two-pole in slots 1–3. Directories were transcribed from door photos (2026-09-20) and checked against each handle rating. One rating is unreadable and is marked with a ? until someone looks with a flashlight.

MAIN PANEL — Square D Homeline 200 A · 40 spaces · basement east wall

1–3	**60**	**GENERATOR subpanel feed (via ATS)**†	2–4	**50**	**EV charger, garage**†
5–7	**40**	**A/C zone 1 condenser**	6	**20**	Microwave
			8	**20**	Kitchen island [DF]
9–11	**30**	**A/C zone 2 condenser**	10	**15**	Family room [AFCI]
			12	**15**	Office / den [AFCI]
13–15	**30**	**Dryer receptacle (unused)**†	14	**20**	Powder room [GFCI]
			16	**15**	Primary bedroom lights [AFCI]†
17	**20**	Kitchen countertop W [DF]	18	**20**	Primary bath [GFCI]
19	**20**	Kitchen countertop E [DF]	20	**15**	Bedroom 2 [AFCI]
21	**20**	Dishwasher [DF]	22	**15**	Stair + upstairs hall
23	**20**	Disposal	24	**20**	Basement outlets
25	**15**	Living room [AFCI]	26	**15**	Basement lights
27	**15**	Dining room [AFCI]	28	**20**	Deck + patio outlets [GFCI]
29	**20**	Garage outlets [GFCI]	30	**15**	Attic fan (none)†
31	**15**	Exterior lights	32		spare
33	**20**	Hall bath [GFCI]	34		spare
35	**15**	Bedrooms 3 + 4 [AFCI]	36		spare
37	**15?**	Smoke/CO (interconnected)†	38		spare
39		spare	40		spare

† 1: feeds the 100 A transfer switch → panel G · 2: 2023, Brightline Electric · 13: dryer is gas; live 240 V circuit to laundry, see Quirks · 16: outlets are on G-8 · 30: fan removed 2019; cable capped in attic · 37: handle rating not legible: verify

GENERATOR PANEL — 12-space subpanel behind the 100 A ATS

1	**20**	Furnace zone 1 (basement)	2	**20**	Laundry (washer + gas dryer)
3	**15**	Furnace zone 2 (attic)	4	**15**	Garage door + garage lights
5	**20**	Sump pump†	6	**15**	Hall + kitchen lights
7	**20**	Refrigerator	8	**20**	Primary bedroom outlets [AFCI]
9	**15**	Water heater power vent†	10	**15**	Radon fan
11	**20**	Kitchen outlets (backed) [DF]†	12	**20**	Basement dehumidifier

† 5: dedicated simplex receptacle · 9: moved here 2025, see Quirks · 11: one outlet left of the sink

Connected load, per leg

Sum of breaker handle ratings on each bus leg: the *most* each leg could draw if every circuit ran at its trip rating at once. It is not a demand load (an NEC 220 calculation would come out far lower), and connected capacity of 2–3× the main is normal. What it is good for is leg balance and headroom. A 2-pole breaker counts its full rating on both legs; slot rows alternate A, B, A, B down the panel. Column = left/right side of the door in amp-poles. In the bar, the tick marks 1× the main and the full bar 3×.

PANEL	MAIN A	LEG A	LEG B	WORST LEG VS MAIN	LEFT / RIGHT	1P / 2P	240 V A	FREE POLES
Panel M	200	440	400	2.20× (380/340 excl. sub-feed)	515 / 325	24 / 5	210	6
Panel G	60	105	110	1.83×	110 / 105	12 / 0	0	0

Panel M's 60 A two-pole feed to the generator panel is counted inside M's legs. Protection: DF = dual-function AFCI/GFCI, AFCI = arc-fault, GFCI = ground-fault. Panel G is rated by its 60 A feed breaker, not its bus.

Connected capacity on panel M is well over the 200 A main, which is normal. The demand estimate below is the number that matters for new loads, and the generator check is the real constraint: panel G's worst leg is connected at about 1.4× the generator's per-leg output, so the furnaces, sump and refrigerator starting together after an outage is the case to watch.

Service demand estimate

The panel table above is connected capacity. This is the code-style demand number: the figure that decides whether the 200 A service needs an upgrade for a new load.

LOAD	BASIS	VA
General lighting & receptacles	2,450 sq ft × 3 VA	7,350
Small-appliance + laundry circuits	4 × 1,500 VA	6,000
Dishwasher	nameplate	1,200
Disposal	nameplate	800
Microwave	typical (no nameplate yet)	1,500
Refrigerator	typical (no nameplate yet)	700
EV charger, 40 A continuous	nameplate	9,600
Water heater power vent	nameplate	300
Garage door opener	typical (no nameplate yet)	500
Radon fan + dehumidifier	typical (no nameplate yet)	900
General loads	first 10 kVA at 100 %, rest at 40 %	28,850 → **17,540**
Heating / cooling	larger of cooling 6,500 or heating 1,600	**6,500**
Calculated demand	24.0 kVA at 240 V	**100 A** = **50 %** of 200 A

Measured Summer peak, both A/C + EV charging 12.4 kW (52 A) · Winter peak 6.1 kW (25 A) · Annual mean 1.4 kW (6 A)

Generator headroom, Panel G. A 19.5 kW unit supplies about 81 A per leg at 240 V. Connected capacity on the backed panel's worst leg is 110 A, **1.4× the generator**: fine only as long as the big loads don't all start together.

Printable door directories are in Appendix C.

4 / Compliance & Inspections

Status as of 2026-10-01. OK current · DUE within a year or now · OVERDUE · UNKNOWN no record found.

ITEM	AUTHORITY	STATUS	LAST	NEXT	NOTES
Smoke + CO alarms	local fire code	OK	2024-01-12	2034-01	Interconnected, replaced 2024 (10-yr sealed batteries). Test monthly.
Radon retest	state guidance	DUE	2018-09-04	2020-09	Retest every 2 years after mitigation and after major HVAC changes; nothing since the post-mitigation test.
Generator permits	building dept	OK	2021-12-03		Electrical + gas permits closed; final sticker inside the transfer switch door.
EV circuit permit	building dept	OK	2023-04-06		Closed.
Backflow (lawn irrigation)	water utility	N/A	?		No irrigation system; utility confirmed no device required.
Deck permit	building dept	UNKNOWN	?		Deckworks said they pulled one; no permit number in any document. Check the building dept's online permit search.
Chimney inspection	NFPA 211 (advisory)	OVERDUE	2018-07-02	2019-07	Fireplace used ~10×/yr; annual inspection recommended.

5 / Quirks & Field Notes

Every entry below cost real time to figure out. Format: what you'll see, why it happens, what to do.

Zone 1 runs all afternoon in July and never makes setpoint

SYMPTOM 1st floor drifts to 78 °F on 90 °F days; condenser runs continuously; supply air feels cool, not cold.

CAUSE Low refrigerant. Topped off 2024-07 (3 lb + UV dye) and again 2026-06 (2 lb). The dye recheck that would have found the leak was never scheduled.

FIX Before paying for a third top-off, have Acme UV-scan the evaporator coil and the line set flare at the condenser. A coil replacement on 2012 equipment is the decision point for replacing the whole zone.

[E Acme invoices 2024-07-16 + 2026-06-23]

Generator exercises, lights flicker at 1 p.m. on Tuesdays

SYMPTOM A brief flicker on generator-backed circuits every Tuesday around 1:00 p.m.

CAUSE That's the weekly exercise cycle. The transfer switch does not transfer during exercise; the flicker is the ATS controller's relay test on this model.

FIX Nothing to fix. If the flicker happens on another day, the generator ran for real: check the controller log.

[GenReady service note 2022-03]

Dead 2-pole breaker labelled "Dryer"

SYMPTOM Main panel M-13/15 is a 30 A 2-pole labelled DRYER, but the dryer is gas.

CAUSE The original 1994 build had an electric-dryer receptacle; the 2018 sellers switched to gas and left the circuit live behind the laundry cabinet.

FIX It's a free 240 V / 30 A circuit already run to the laundry room. Reuse it (heat-pump water heater, a future mini-split) before pulling a new home run.

[N panel photo 2026-09; inspection 2018]

Attic fan breaker with no fan

SYMPTOM M-30 (15 A) labelled ATTIC FAN; there is no attic fan.

CAUSE Northwind removed the gable fan during the 2019 air-seal (it was depressurizing the attic and pulling conditioned air); the cable is capped in a junction box on the attic floor.

FIX Leave it labelled "spare (capped in attic)". Do not let anyone re-install a powered attic fan over the R-49.

[S Northwind scope 2019-10]

Sump runs every 4 minutes in April

SYMPTOM Primary pump cycles on short intervals for a few weeks each spring; in dry months it barely runs.

CAUSE High seasonal water table, not a fault. The backup pump has only ever run during the 2021 float failure.

FIX Pour a bucket into the pit each October to exercise both pumps; listen for the backup alarm on the battery test button.

[M 2021-04 + field notes]

Hot water dies in a power outage (before 2025)

SYMPTOM During the 2023 ice storm the generator ran but there was no hot water.

CAUSE The previous water heater's power-vent blower was plugged into a basement outlet on the main panel, not the generator panel.

FIX Fixed in 2025: the new heater's receptacle was moved to generator circuit G-9. Any future replacement: keep the vent on G-9.

[E Dry Basin 2025-02 work order]

6 / Service Providers

PROVIDER	TRADE	CONTACT	WORK HERE	WHEN
Deckworks Outdoor	Carpentry	555-0176	Deck rebuild 2020	2020
Sootbusters Chimney	Chimney	555-0122	Sweep + inspection 2018	2018
Brightline Electric	Electrical	555-0129	EV charger circuit 2023	2023
GenReady Power Systems	Generator	555-0163	Generator install 2021; annual service every March	2021–26
Clearview Window & Glass	Glass	555-0190	2 IGUs 2023; quote for 6 more on request	2023
Acme Heating & Cooling	HVAC	555-0117 · service@acme-hvac.example	Fall tune-ups since 2018; zone 2 replacement 2020; zone 1 refrigerant 2024 + 2026	2018–26
Greenway Hardscapes	Hardscape	555-0134	Paver patio 2024; warranty claim pending	2024–26
Northwind Insulation	Insulation	555-0155	Attic air-seal + R-49, 2019	2019
Maple Valley Pest	Pest	555-0149	Carpenter ants 2021	2021
Dry Basin Plumbing	Plumbing	555-0142	Sump + backup 2021; water heater 2025; backup battery	2021–25
Riverbend Radon Services	Radon	555-0108	Mitigation 2018	2018
Summit Roofing Co.	Roofing	555-0181	2019 tear-off. Also works next door: check the address on every invoice	2019

7 / Warranties

Status as of 2026-10-01: ACTIVE more than a year left, EXPIRING under a year, EXPIRED, UNKNOWN term or start date not found.

ITEM	WARRANTOR	TERM	START	EXPIRES	STATUS	DOC
Zone 2 labor	Acme Heating & Cooling	2 yr labor	2020-04-22	2022-04-22	EXPIRED	ON FILE
Radon system	Riverbend Radon Services	fan 5 yr; system transferable	2018-08-20	2023-08-20	EXPIRED	ON FILE
Deck workmanship	Deckworks Outdoor	5 yr	2020-08-10	2025-08-10	EXPIRED	PARTIAL
Water heater labor	Dry Basin Plumbing	1 yr	2025-02-10	2026-02-10	EXPIRED	ON FILE
Generator	Generac (limited)	5 yr	2021-11-15	2026-11-15	EXPIRING	ON FILE
Patio settling	Greenway Hardscapes	3 yr installer, settling > 1/2 in	2024-04-30	2027-04-30	EXPIRING	ON FILE
Roof workmanship	Summit Roofing Co.	10 yr workmanship	2019-05-14	2029-05-14	ACTIVE	ON FILE
Zone 2 furnace + A/C parts	Carrier, registered	10 yr parts	2020-04-22	2030-04-22	ACTIVE	ON FILE
Water heater tank	Bradford White	6 yr tank	2025-02-10	2031-02-10	ACTIVE	ON FILE
Window glass (2 IGUs)	Clearview Window & Glass	10 yr seal	2023-06-19	2033-06-19	ACTIVE	MISSING
Deck boards	board manufacturer	25 yr fade + stain	2020-08-10	2045-08-10	ACTIVE	PARTIAL
Roof system (manufacturer)	shingle manufacturer, via contractor	enhanced system, if registered	2019-05-14	?	UNKNOWN	MISSING

8 / Invoice Ledger

DATE	PROJECT	VENDOR	DOCUMENTED	PAID (TRACED)	PAPERWORK
2018-08-20	Radon mitigation	Riverbend Radon Services	$1,450	$1,450	ON FILE
2019-05-14	Roof tear-off and re-shingle	Summit Roofing Co.	$14,900	$14,900	PARTIAL
2019-10-03	Attic air-seal + blown cellulose to R-49	Northwind Insulation	$3,280	$3,280	ON FILE
2020-04-22	Zone 2 furnace + A/C replacement (attic)	Acme Heating & Cooling	$9,850	$9,850	ON FILE
2020-08-10	Deck rebuild, composite	Deckworks Outdoor	$18,400	$17,900	PARTIAL
2021-05-03	Sump pump replacement + battery backup	Dry Basin Plumbing	$1,180	$1,180	ON FILE
2021-11-15	Standby generator + transfer switch + subpanel	GenReady Power Systems	$13,600	$13,600	ON FILE
2022-09-12	Gutters + step flashing (NOT this house)	Summit Roofing Co.	$6,850	—	ON FILE
2023-03-28	EV charger circuit (50 A)	Brightline Electric	$1,650	$1,650	ON FILE
2023-06-19	Dining-room window glass (2 sashes)	Clearview Window & Glass	$940	$940	PARTIAL
2024-04-30	Paver patio (380 sq ft)	Greenway Hardscapes	$11,200	$11,200	ON FILE
2025-02-10	Water heater replacement	Dry Basin Plumbing	$3,450	$3,450	ON FILE
2025-09-08	Interior paint, 2nd floor	Hometown Painting	$5,300	$2,650	ASK ALEX
	12 projects (excl. disputed)		**$85,200**	**$82,050**	
	Best estimate (paid where traced, else documented)			**$82,050**	

8.1. Individual invoices & receipts on file

DATE	VENDOR	WHAT	AMOUNT	WHERE FILED
2018-07-02	Sootbusters Chimney	Sweep + level-2 inspection, family-room fireplace; crown sealed	$289	S 0101
2018-08-20	Riverbend Radon Services	Radon mitigation system	$1,450	S 0112
2018-10-15	Acme Heating & Cooling	Fall tune-up, both zones	$189	E
2019-05-14	Summit Roofing Co.	Roof tear-off + re-shingle	$14,900	S 0187
2019-10-03	Northwind Insulation	Attic air-seal + R-49 cellulose	$3,280	S 0203
2020-03-11	Acme Heating & Cooling	Zone 2 no-heat diagnostic: cracked heat exchanger, red-tagged	$129	E
2020-04-22	Acme Heating & Cooling	Zone 2 furnace + A/C replacement	$9,850	S 0221
2020-08-10	Deckworks Outdoor	Deck rebuild (deposit + final)	$17,900	B
2021-04-29	Dry Basin Plumbing	Emergency call: sump float stuck, pit near full	$225	M + S 0243
2021-05-03	Dry Basin Plumbing	Sump pump + battery backup	$1,180	S 0244
2021-06-14	Maple Valley Pest	Carpenter ants, garage sill plate; treatment + 1 follow-up	$310	E
2021-11-15	GenReady Power Systems	Standby generator, ATS, subpanel, gas upsizing	$13,600	S 0281
2022-03-02	GenReady Power Systems	Annual service: oil, filter, plugs	$295	E
2022-09-12	Summit Roofing Co.	Gutters + flashing at 71 Maple Ct, NOT this house (excluded)	$6,850	S 0310
2022-10-05	Acme Heating & Cooling	Fall tune-up, both zones	$199	E
2023-03-02	GenReady Power Systems	Annual service + battery	$412	E
2023-03-28	Brightline Electric	EV charger circuit + permit	$1,650	S 0331
2023-06-19	Clearview Window & Glass	2 insulated-glass units, dining room	$940	B
2023-10-11	Acme Heating & Cooling	Fall tune-up, both zones	$209	E
2024-03-06	GenReady Power Systems	Annual service	$310	E
2024-04-30	Greenway Hardscapes	Paver patio	$11,200	S 0367
2024-07-16	Acme Heating & Cooling	Zone 1 no-cool: +3 lb R-410A + UV dye; tech note “recheck with UV in 30 days”	$485	E
2024-10-09	Acme Heating & Cooling	Fall tune-up, both zones (dye recheck NOT done)	$219	E
2025-02-10	Dry Basin Plumbing	Water heater replacement	$3,450	S 0402
2025-03-05	GenReady Power Systems	Annual service	$325	E
2025-09-08	Hometown Painting	2nd-floor paint, deposit	$2,650	B
2025-10-20	Dry Basin Plumbing	Sump backup battery	$235	E
2026-03-04	GenReady Power Systems	Annual service; exercise moved to Tuesdays	$340	E
2026-06-23	Acme Heating & Cooling	Zone 1 low charge again: +2 lb R-410A; coil leak suspected, quote $2,900 for evaporator coil	$395	E

9 / Open Issues

Known problems and unknowns as of 2026-10-01, safety first.

KIND	ISSUE
SAFETY	Radon retest overdue since 2020; the 2020 HVAC replacement changed basement pressure.
SAFETY	Panel M-37 (smoke/CO) handle rating unreadable: verify it's 15 A, not 20 A on 14 AWG.
MECHANICAL	Zone 1 A/C leaking refrigerant (5 lb over two summers); leak not located.
MECHANICAL	Zone 1 furnace + A/C are 2012 equipment; the seller sheet's capacities were wrong (nameplate: 80k / 3.5 t).
ENVELOPE	Six original 1994 window units show early seal fogging.
ENVELOPE	Patio low spot by the NE downspout; under installer warranty until 2027-04-30.
PAPERWORK	Roof manufacturer warranty registration, deck change order + permit number, Clearview invoice, paint balance.
COMFORT	Bedroom 4 (over the garage) runs 3–4 °F warmer than the rest of zone 2 in summer.
PLANNED	Zone 1 replacement decision (see the project queue).

9.1. Project queue

Zone 1 HVAC: repair or replace

WHY Leaking 2012 system, two top-offs, a $2,900 coil quote on the table.

PLAN UV leak search first. If it's the coil, get a full-replacement quote sized from a load calc (the seller's 4-ton claim was wrong; it's 3.5 t now and may only need 3). Consider a heat pump: the dead 30 A dryer circuit and 200 A demand headroom both allow it.

Re-glaze or replace the remaining six windows

WHY Early seal failure on six 1994 units.

PLAN Price IGU-only replacement (Clearview) against full inserts. IGU-only keeps the frames and costs ~30 % as much.

Bedroom 4 comfort

WHY Room over the garage, warmer in summer.

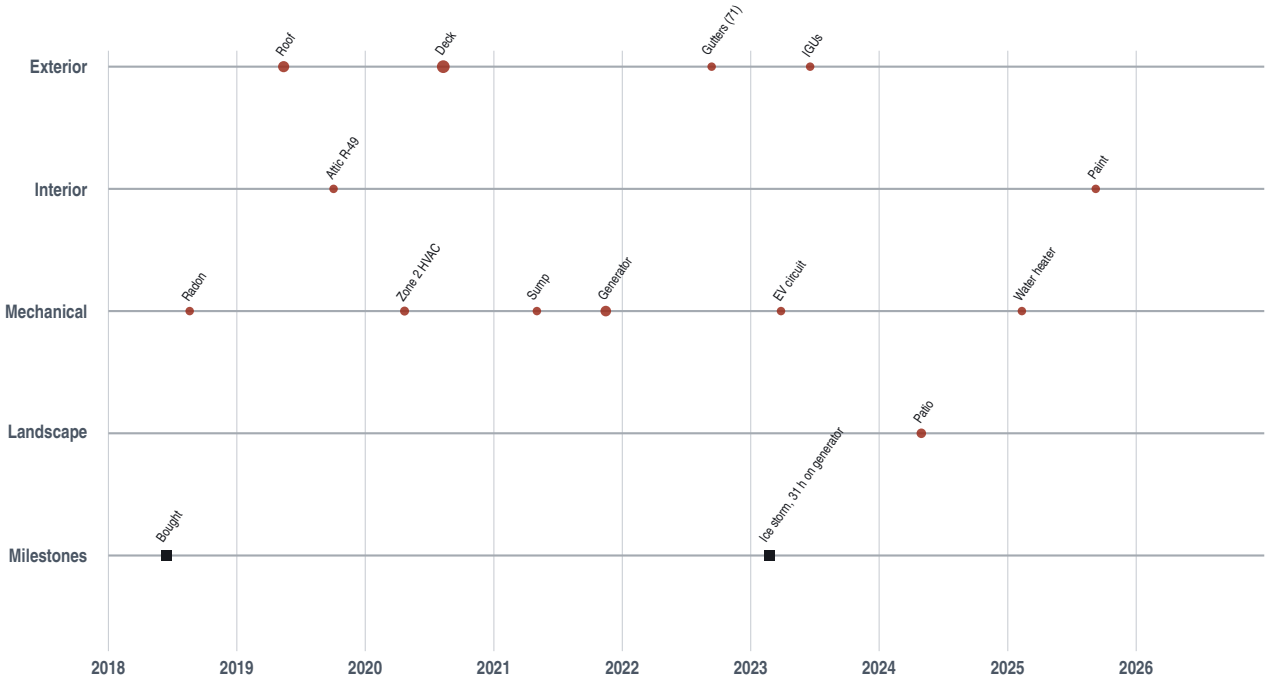
PLAN Check the duct run in the garage ceiling for disconnection first; then a balancing damper. Mini-split only if both fail.

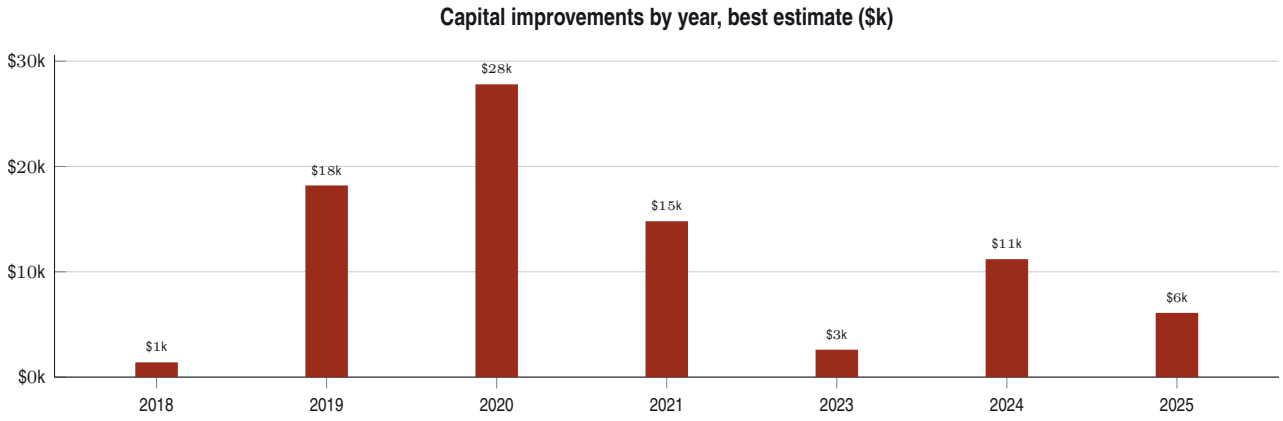
10 / Maintenance Calendar

WHEN	TASK	WHY / NOTES
Monthly	Test smoke/CO alarms; check the radon manometer (liquid offset, not level)	A level manometer = fan stopped.
Every 90 days	Zone 2 filter, 20×20×1 at the attic return	Reachable through the bedroom 4 closet hatch.
Every 6 months	Zone 1 media filter, 16×25×4	Basement cabinet next to the furnace.
March	Generator annual service (GenReady)	Oil, filter, plugs, battery load test.
April	Clear the condenser fins + level the pads; confirm both A/C zones before the first hot week	Zone 1 is the known leaker.
May	Gutters + downspout extensions; check the patio for settling	Patio warranty claims need dated photos.
October	HVAC fall tune-up, both zones (Acme)	Ask for the zone 1 UV dye check until the leak is found.
October	Bucket test both sump pumps; press the backup's battery test	The 2021 failure was a stuck float.
November	Shut and drain the hose bibs from inside (frost-free, but disconnect hoses)	A frost-free bib with a hose attached still splits.
Every 2 years	Radon retest (short-term kit, basement, closed-house conditions)	Overdue since 2020.
Every 5 years	Sump backup battery	Last 2025-10.
Annually	Chimney inspection before the first fire	Last 2018.

A / Eight Years in One Line

From 2018 through 2026-10-01, **$82,050** of documented capital improvements.





B / Improvements

Radon mitigation

VENDOR Riverbend Radon Services COMPLETED 2018-08-20 COST $1,450 WARRANTY fan 5 yr · system transferable PAPERWORK ON FILE

- Sub-slab depressurization, single suction point in the basement SE corner
- Fan in the garage attic, stack through the garage roof
- Pre-mitigation 6.2 pCi/L (home-sale test) → post 0.9 pCi/L

[On file: contract + post-test report · E 2018-08-21; S 0112]

Roof tear-off and re-shingle

VENDOR Summit Roofing Co. COMPLETED 2019-05-14 COST $14,900 WARRANTY 10 yr workmanship · mfr system (unconfirmed) PAPERWORK PARTIAL

- Tear-off to the deck, 2 layers; replaced 6 sheets of rotted OSB at the rear eave
- Synthetic underlayment; ice & water shield at eaves, valleys and the garage-wall step flashing
- Architectural shingles + ridge vent; new pipe boots

Contract promises the manufacturer's enhanced system warranty, which only exists if the contractor registers it. No registration certificate in any source.



After tear-off

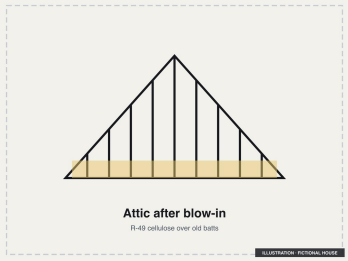
[On file: signed contract, final invoice; no warranty registration · E 2019-04-02 contract; S 0187]

Attic air-seal + blown cellulose to R-49

VENDOR Northwind Insulation COMPLETED 2019-10-03 COST $3,280 WARRANTY — PAPERWORK ON FILE

- Air-sealed top plates, plumbing + wiring penetrations, the bath-fan housings
- Baffles in every soffit bay
- Blown cellulose to R-49 over existing R-19 batts
- Removed the powered gable fan (cable capped)

Qualified for a utility rebate ($0.50/sq ft attic, ~$600) that was never filed. The rebate window has closed.



After the blow-in

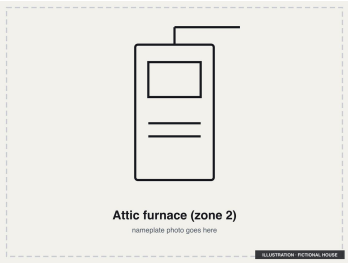
[On file: invoice + depth certificate stapled in the attic · S 0203]

Zone 2 furnace + A/C replacement (attic)

VENDOR Acme Heating & Cooling COMPLETED 2020-04-22 COST $9,850 WARRANTY 10 yr parts (registered) · 2 yr labor PAPERWORK ON FILE

- Carrier 59SC5A060 furnace (60k) in the attic, new secondary pan + float cutoff
- 24ACC624 condenser (2 t) on a new pad

- New line set; old one had a kinked suction line
- Replaced a cracked heat exchanger found at the 2020-03 diagnostic



Zone 2 furnace

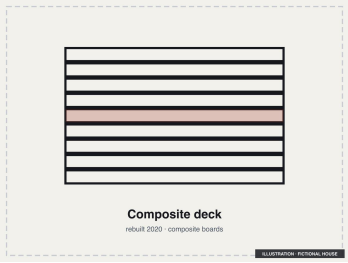
[On file: proposal, invoice, registration confirmation · E 2020-03-11 diagnostic; E 2020-04-23 registration]

Deck rebuild, composite

VENDOR Deckworks Outdoor COMPLETED 2020-08-10 COST $18,400 · paid $17,900 WARRANTY 5 yr workmanship · 25 yr boards PAPERWORK PARTIAL

- Demo of the original cedar deck; existing footings inspected and retained ($500 credit)
- New PT framing with joist tape; hidden-fastener composite boards
- Aluminum rail; stair to the yard relocated to the east side

Only the deposit and final card payment are traced; the signed change order for the stair relocation isn't on file.



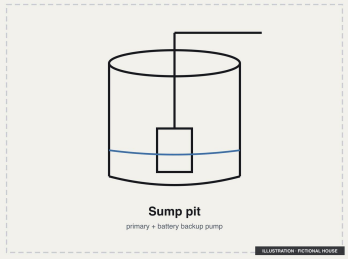
Composite deck

[On file: proposal, two card payments · E 2020-06-30 proposal; B 2020-06 + 2020-08]

Sump pump replacement + battery backup

VENDOR Dry Basin Plumbing COMPLETED 2021-05-03 COST $1,180 WARRANTY — PAPERWORK ON FILE

- Zoeller M53 primary (old pump's float switch stuck, 2021-04-29 emergency visit)
- Basement Watchdog battery backup with alarm + 2nd check valve
- Dedicated simplex receptacle (later moved to generator panel G-5)



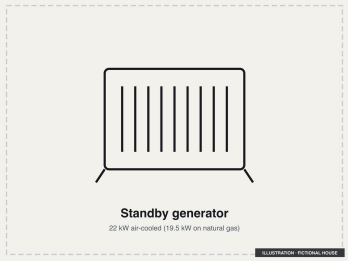
Sump pit, both pumps

[On file: invoice · S 0244; M 2021-04-29]

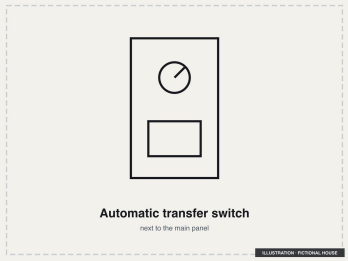
Standby generator + transfer switch + subpanel

VENDOR GenReady Power Systems COMPLETED 2021-11-15 COST $13,600 WARRANTY 5 yr limited (mfr) · 1 yr install PAPERWORK ON FILE

- Generac Guardian 22 kW on a composite pad, north side
- 100 A automatic transfer switch + 12-space generator subpanel; 12 circuits moved from the main panel
- Gas line upsized from the meter (separate permit)
- Electrical + gas permits closed 2021-12-03



Generator



Transfer switch
[On file: contract, permits, final inspection sticker · E 2021-09 contract; S 0281]

Gutters + step flashing (NOT this house) EXCLUDED

VENDOR Summit Roofing Co. COMPLETED 2022-09-12 COST $6,850 WARRANTY — PAPERWORK ON FILE

Arrived in the same inbox as the 2019 roof and was filed here for three years. The invoice is addressed to 71 Maple Court: Summit did that house the same week. Nothing was done here in 2022; the 2019 roof job already included the step flashing.

[E 2022-09-14; S 0310 (address line)]

EV charger circuit (50 A)

VENDOR Brightline Electric COMPLETED 2023-03-28 COST $1,650 WARRANTY — PAPERWORK ON FILE

- 50 A 2-pole breaker M-2/4, 6 AWG copper to the garage
- Hardwired 40 A wall unit (owner-supplied)
- Load calc on the permit: 200 A service OK

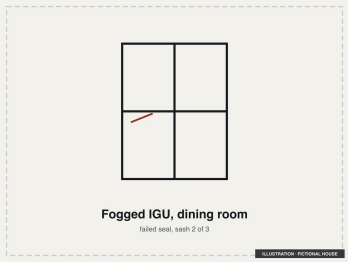
[On file: invoice + permit · S 0331]

Dining-room window glass (2 sashes)

VENDOR Clearview Window & Glass COMPLETED 2023-06-19 COST $940 WARRANTY 10 yr seal (glass only) PAPERWORK PARTIAL

- Replaced two fogged insulated-glass units in the original sashes
- Low-E argon units; frames kept

Paid by card; the emailed invoice never arrived.



Fogged glass before

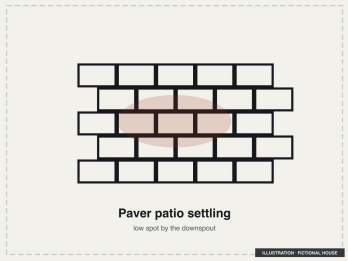
[On file: card statement only · B 2023-06; M 2023-06-12]

Paver patio (380 sq ft)

VENDOR Greenway Hardscapes COMPLETED 2024-04-30 COST $11,200 WARRANTY 3 yr installer (to 2027-04-30) PAPERWORK ON FILE

- Excavated 10 in; 6 in compacted base, 1 in bedding sand
- Concrete pavers, soldier-course border, polymeric sand
- Downspout extension re-routed under the patio in 4 in PVC

The 3-year installer warranty explicitly covers settling over 1/2 in. The low spot by the NE downspout (2026-05) qualifies.



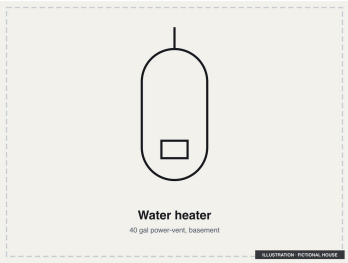
Settling, 2026-05

[On file: contract with warranty clause · E 2024-03 contract]

Water heater replacement

VENDOR Dry Basin Plumbing COMPLETED 2025-02-10 COST $3,450 WARRANTY 6 yr tank · 1 yr labor PAPERWORK ON FILE

- Bradford White 40-gal gas power-vent (replaced a leaking 2009 tank)
- Receptacle moved to generator circuit G-9
- Expansion tank + new gas flex + drain pan



New heater

[On file: invoice + registration · S 0402]

Interior paint, 2nd floor

VENDOR Hometown Painting COMPLETED 2025-09-08 COST $5,300 · paid $2,650 WARRANTY — PAPERWORK ASK ALEX

- 4 bedrooms + hall + stair, walls and ceilings
- Drywall repair at the stair stringer crack

Only the 50 % deposit is traced. The balance was likely paid by check.

[On file: estimate; deposit on card · E 2025-08 estimate; B 2025-08]

C / Printable Panel Directories

One letter page per panel, generated from the same directory data as Chapter 3. Print at 100 %, trim, and tape inside each panel door.

MAIN PANEL

Square D Homeline 200 A · 40 spaces · basement east wall

17 Maple Court · surveyed 2026-09-20 · amps from each handle · shaded rows = 2-pole (240 V) · centre strip = bus leg

240 V **GENERATOR subpanel feed (via ATS)** *feeds the 100 A transfer switch → panel G*	**60**	1 3	A B	2 4	**50**	**EV charger, garage** 240 V *2023, Brightline Electric*
240 V **A/C zone 1 condenser**	**40**	5 7	A	6	**20**	Microwave
			B	8	**20**	Kitchen island DF
240 V **A/C zone 2 condenser**	**30**	9 11	A	10	**15**	Family room AFCI
			B	12	**15**	Office / den AFCI
240 V **Dryer receptacle (unused)** *dryer is gas; live 240 V circuit to laundry, see Quirks*	**30**	13 15	A	14	**20**	Powder room GFCI
			B	16	**15**	Primary bedroom lights AFCI *outlets are on G-8*
DF Kitchen countertop W	**20**	17	A	18	**20**	Primary bath GFCI
DF Kitchen countertop E	**20**	19	B	20	**15**	Bedroom 2 AFCI
DF Dishwasher	**20**	21	A	22	**15**	Stair + upstairs hall
Disposal	**20**	23	B	24	**20**	Basement outlets
AFCI Living room	**15**	25	A	26	**15**	Basement lights
AFCI Dining room	**15**	27	B	28	**20**	Deck + patio outlets GFCI
GFCI Garage outlets	**20**	29	A	30	**15**	Attic fan (none) *fan removed 2019; cable capped in attic*
Exterior lights	**15**	31	B	32		spare
GFCI Hall bath	**20**	33	A	34		spare
AFCI Bedrooms 3 + 4	**15**	35	B	36		spare
Smoke/CO (interconnected) *handle rating not legible: verify*	**15?**	37	A	38		spare
spare		39	B	40		spare

Connected load leg A 440 A · leg B 400 A against a 200 A main (2.20× worst leg) · 24 single-pole, 5 two-pole · 6 free poles · incl. the 60 A sub-feed

Print at 100 % on letter; trim and tape inside the panel door. Pencil corrections in the blank rows, then fix them in the book data.

GENERATOR PANEL

12-space subpanel behind the 100 A ATS

17 Maple Court · surveyed 2026-09-20 · amps from each handle · shaded rows = 2-pole (240 V) · centre strip = bus leg

EVERY CIRCUIT ON THIS PANEL IS GENERATOR-BACKED (22 kW standby via automatic transfer switch)

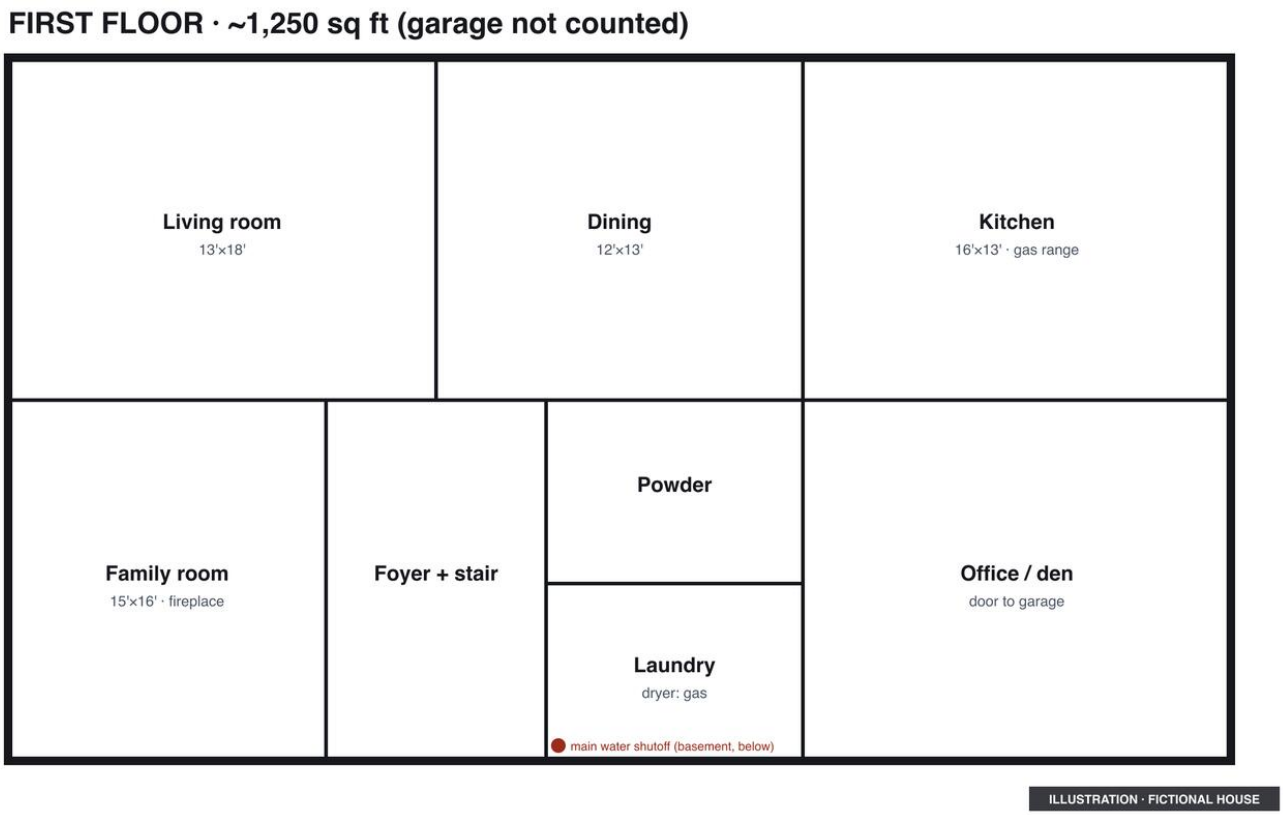
Furnace zone 1 (basement)	**20**	1	A	2	**20**	Laundry (washer + gas dryer)
Furnace zone 2 (attic)	**15**	3	B	4	**15**	Garage door + garage lights
Sump pump *dedicated simplex receptacle*	**20**	5	A	6	**15**	Hall + kitchen lights
Refrigerator	**20**	7	B	8	**20**	Primary bedroom outlets AFCI
Water heater power vent *moved here 2025, see Quirks*	**15**	9	A	10	**15**	Radon fan
DF Kitchen outlets (backed) *one outlet left of the sink*	**20**	11	B	12	**20**	Basement dehumidifier

Connected load leg A 105 A · leg B 110 A against a 60 A feed (1.83× worst leg) · 12 single-pole, 0 two-pole · 0 free poles

Print at 100 % on letter; trim and tape inside the panel door. Pencil corrections in the blank rows, then fix them in the book data.

D / Drawings & Plans

No permit drawings survive from the 1994 build; these sketches were drawn from the 2018 inspection report and a tape measure. They show room names, not circuits; the panel chapter is the circuit map.



First floor. Main water shutoff is directly below the laundry, on the basement north wall.

SECOND FLOOR · ~1,200 sq ft

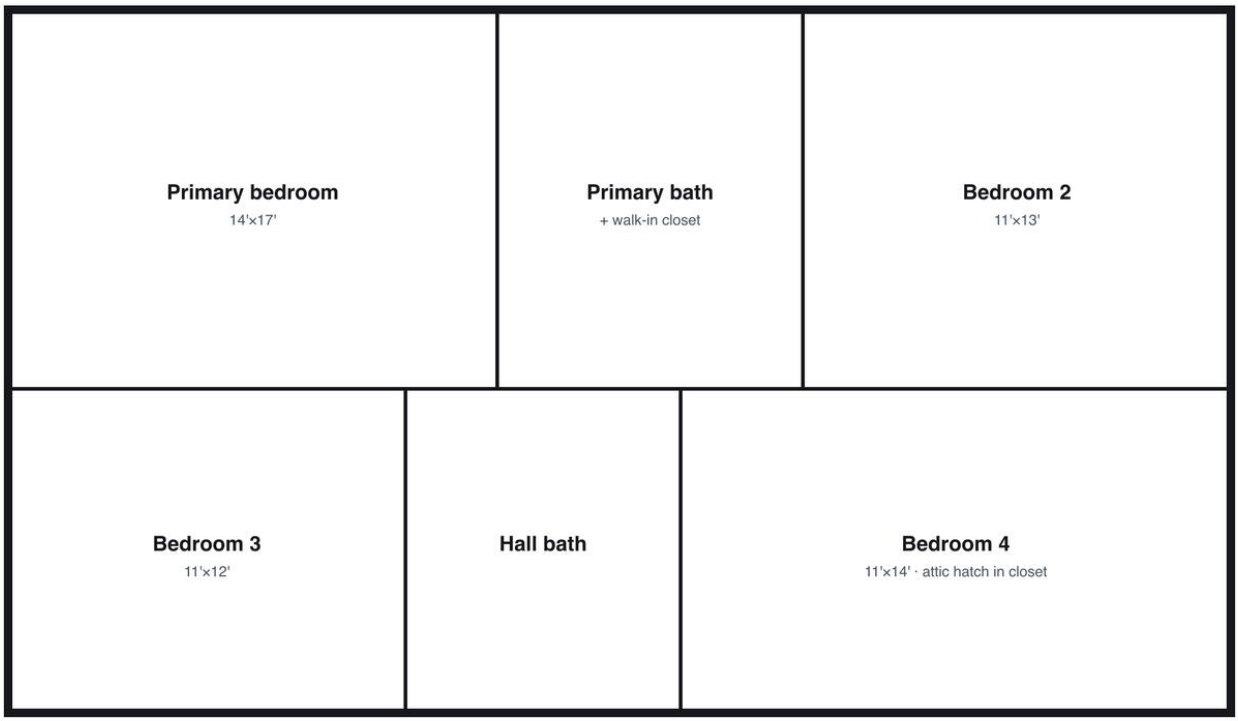


ILLUSTRATION · FICTIONAL HOUSE

Second floor. Zone 2 attic access is the hatch in the bedroom 4 closet.

E / Equipment Manuals

7 manufacturer documents, filed in `manuals/` and shared alongside the photo index. Match: EXACT this model · FAMILY covers the series · LIKELY · UNCONFIRMED read the nameplate.

SYSTEM	EQUIPMENT	MAKER / MODEL	DOCUMENT	PP	MATCH
Electrical	Standby generator	Generac Guardian 22 kW	owner's manual	60	LIKELY
Electrical	Transfer switch	Generac 100 A ATS	installation	28	UNCONFIRMED
HVAC	Both condensers	Carrier 24ACC6	installation	24	FAMILY
HVAC	Zone 1 furnace	Carrier 59TP5A080	owner's manual	16	FAMILY
HVAC	Zone 2 furnace	Carrier 59SC5A060	installation	52	EXACT
Plumbing	Sump pump	Zoeller M53	owner's manual	8	EXACT
Plumbing	Water heater	Bradford White 40-gal power vent	installation	44	FAMILY

Nameplate walk-through

Model numbers still needed. One pass through the house with a phone camera closes these out:

ITEM	WHERE TO LOOK	WHY
Zone 1 condenser	east side yard, left unit, label on the back corner	confirm 3.5 t and the factory charge for the leak math
Main panel catalog label	inside the door, lower left	confirm copper bus and the max-breaker-space count
Dishwasher	inside the door edge	model for the manuals table
Garage door opener	on the motor head	model + horsepower (it's on the generator)
Breaker M-37	main panel, left column, near the bottom	handle rating unreadable in the photo